

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and Environmental Engineering) SEMESTER - 6 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and Environmental Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCEC603 - INDUSTRIAL WASTE TREATMENT

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 What is an essential parameter in calculating WQI?
 1. Biological Oxygen Demand (BOD)
 2. Population density
 3. Industrial growth rate
 4. Sludge generation rate
- 2 What is the purpose of river standards?
 1. Define minimum flow rates
 2. Protect aquatic ecosystems
 3. Encourage industrial use
 4. Promote tourism
- 3 What is the main challenge in operating CETPs.
 1. Lack of manpower
 2. High energy consumption
 3. Maintenance issues
 4. Lack of raw materials
- 4 What is a characteristic of sugar industry wastewater
 1. High organic load
 2. High ammonia content
 3. High fluoride levels
 4. High chromium content
- 5 What is a key feature of tannery wastewater treatment
 1. Chromium removal
 2. Pathogen reduction
 3. Oil skimming
 4. Phosphate addition

- 6 What is a major use of water in industries
 1. Irrigation
 2. Cooling
 3. Recreation
 4. Drinking water
 - 7 What is a significant factor in wastewater flow monitoring
 1. Volume per time
 2. Treatment plant cost
 3. Soil type
 4. Land use
 - 8 What does waste segregation achieve
 1. Simplifies treatment
 2. Reduces treatment cost
 3. Avoids regulatory issues
 4. All of the above
 - 9 What is the main feature of Zero Liquid Discharge (ZLD)
 1. No wastewater discharge
 2. Partial waste discharge
 3. Complete toxicity removal
 4. Open disposal
 - 10 What is a benefit of waste reuse
 1. Reduces raw material cost
 2. Increases energy use
 3. Creates more waste
 4. Increases treatment complexity
 - 11 What does an online equalization tank provide
 1. Real-time flow balancing
 2. Long-term storage
 3. Waste segregation
 4. Sedimentation
 - 12 What is a key feature of treatment for cyanide removal?
 1. Oxidation
 2. Precipitation
 3. Neutralization
 4. Sedimentation
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- Q2. Solve the following.
- A) Describe the effects of industrial waste water on rivers. 6
- B) Explain proportioning of industrial waste. 6
- Q3. Solve the following.
- A) Differentiate between industrial waste water and domestic waste water. 6
- B) Write a short note on grab sampling and composite sampling. 6
- Q4. Solve Any Two of the following.
- A) Explain phenomenon of equalization of water . 6

- B) Explain recycle, Re-use, and by product recovery 6
- C) Describe the characteristics of dairy industrial waste water and develop treatment flow sheet. 6
- Q5. Solve Any Two of the following.
- A) Explain water pollution control act . 6
- B) Write a short note on bioremediation. 6
- C) Explain ISO 14000 and it's significance for industries. 6
- Q6. Solve Any Two of the following.
- A) Write a short note on environmental bio technology. 6
- B) Explain treatability index. 6
- C) Explain environmental management plan. 6

*** End ***